

A serene sunset scene over a pond. The sky transitions from a deep blue to a warm orange glow near the horizon. Silhouetted trees line the background and the right side of the frame. In the center-left of the pond, a fountain sprays water upwards, illuminated from below by a bright yellow light that reflects on the water's surface. The overall mood is calm and peaceful.

Stress

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Inspo

- GA tech news article
- Garmin

Problem?



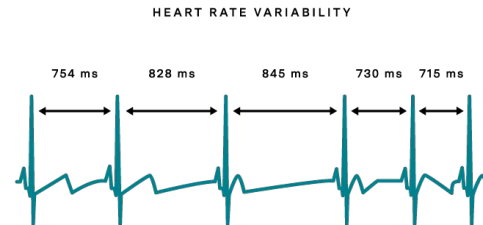
Solution.

Stress Metrics

- Heart Rate (HR) in bpm
 - Baseline HR
 - HR delta (current HR - baseline HR)
 - HRV change percent from baseline
- RR Interval (RR) in milliseconds between beats
 - Context flags:
 - Resting
 - Exercise like
 - Recovery window
 - Root Mean Square of the Successive Differences (RMSSD) from RR intervals

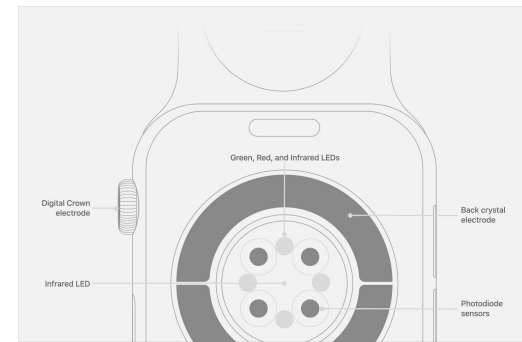
- Sources

- <https://www.garmin.com/en-US/garmin-technology/health-science/stress-tracking/>
- <https://www.whoop.com/thelocker/heart-rate-variability-hrv>
- <https://support.garmin.com/en-US/?faq=PB1duL5p6V64lQwhNvcRK9&productID=730659&tab=topics>



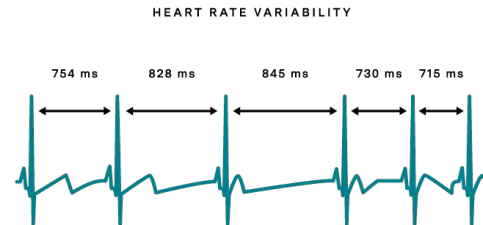
Stress Measurements

- Sensors measure *skin, sweat, heart beat*
- Skin (Apple, Garmin)
 - **For HRV**
 - Green light (most similar to EKG/ECG) shines on your skin and measures blood flow
 - More blood = more absorption, less reflection
 - Photodiode measures reflected light
 - Peak reflection → less blood in the capillary at that moment
 - Trough reflection → more blood during heartbeat
 - Pulse waveform is reconstructed from these peaks and troughs showing the RR intervals
 - **For blood oxygen**
 - Red and infrared light estimates blood oxygen or % of oxygen saturated hemoglobin in blood
 - The light goes into your skin (to capillaries) > some light is absorbed by blood > some reflects back
 - Oxygenated → absorbed more infrared, less red
 - Deoxygenated → absorbed more red, less infrared
 - Light reflected is measured by photodiodes
- Sources
 - https://youtu.be/_5Nh2wnChpc?si=MmsGdKONMN85K5J8
 - <https://support.apple.com/en-us/120277>



Stress Estimator

- Stress likelihood increases when:
 - HR is above baseline
 - RMSSD is below baseline
 - Deviation is sustained overtime
- False positive anticipation
 - When user is exercising
- Output to user
 - Stress estimate: 0% - 100%
 - Confidence level: low, medium, high
 - Safety statement to remind the user this is ***not*** a medical diagnosis



Grounding techniques

- 5,4,3,2,1
 - 5 things you see, ..., 2 things you smell, 1 thing you taste
- 1 x 30s
 - Focus on 1 stimuli for 30s
- **4x4 aka box breathing**
 - **Inhale 4s, hold 4s, exhale 4s, hold 4s**
- **Resonate breathing**
 - **5s inhale, 5s exhale**
- Naming (most friendly for echolalia people)
 - Name categories (favorite songs)
 - Name facts (date, time, location)
 - Recite (periodic table, times table, etc)
- “Touch grass”
 - Sitting on the ground/flat surface
 - Pace and count steps
 - Basically lower the center of gravity

Demo!

Name contenders

- Stressa Inte
- 걱정하지 마 geok jeong hajima
- 没事 mei shi